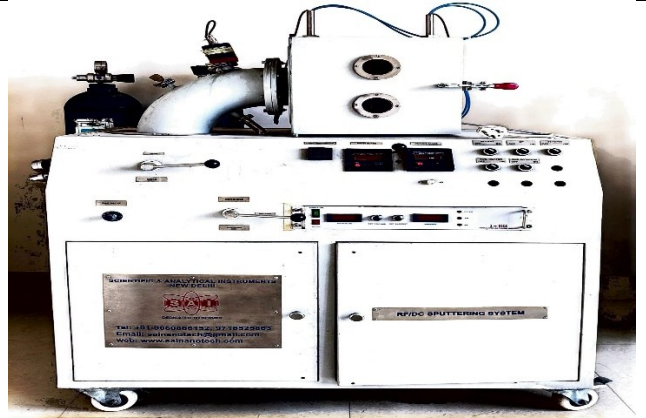


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Equipment Overview

RF/DC Sputtering Setup:

An **RF/DC Sputtering Machine** is an industrial or laboratory physical vapour deposition (PVD) system designed to apply precision thin-film coatings onto a substrate. By combining both **Direct Current (DC)** and **Radio Frequency (RF)** power generators into a single vacuum chamber tool, this machine allows operators to switch between depositing conducting metals and insulating dielectrics without changing hardware setups.



Spin Coater Setup:

A **Spin Coater** is a precision laboratory instrument used to deposit uniform, ultra-thin films of liquid materials onto flat substrates. By utilizing centrifugal force, it spreads a fluid across a spinning platform, achieving highly controllable film thicknesses ranging from a few nanometers to several micrometers.



Magnetic Stirrer:

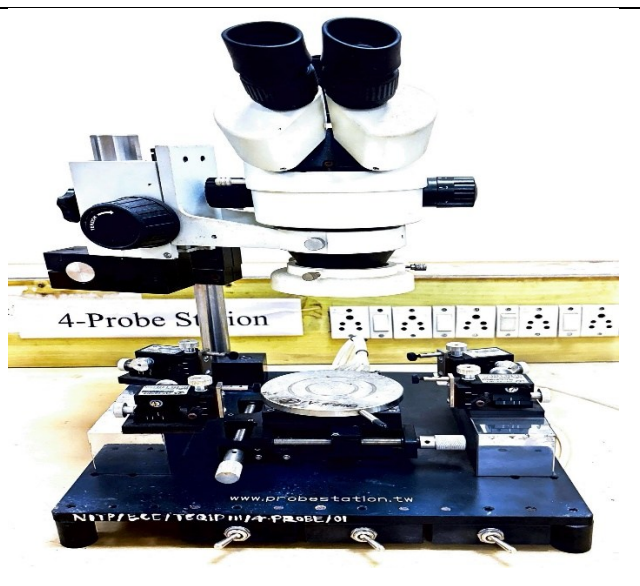
A **Magnetic Stirrer** (frequently paired with a heating element as a **Magnetic Stirrer Hotplate**) is a fundamental laboratory instrument used to mix low-viscosity liquids or dissolve solutes in a solvent. It uses a rotating magnetic field to cause a submerged magnetic stir bar (commonly called a "flea") to spin rapidly within the vessel, creating a continuous, hands-free vortex.



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Four Probe station:

A **Four-Probe Station** (or **Four-Point Probe Station**) is an integrated mechanical and electrical workspace used to probe and characterize materials, semiconductors, thin films, and microelectronic devices. While a standalone four-probe setup usually features a fixed, single-head assembly for bulk materials or flat wafers, a *probe station* is a versatile platform equipped with independent, micro-positionable arms [cascademicrotech.com]. This allows engineers to place microscopic pins onto specific contacts, traces, or pads on an integrated circuit (IC) or device wafer



Probe Sonicator:

A **Probe Sonicator** (frequently called an **Ultrasonic Homogenizer** or **Ultrasonic Cell Disruptor**) is a high-intensity laboratory instrument used to agitate particles in a liquid sample using high-frequency sound waves. Unlike ultrasonic cleaning baths that disperse sound waves throughout a large tank, a probe sonicator directs intense ultrasonic energy straight into a localized sample via a submerged metal probe.

This process generates extreme physical forces capable of breaking down cellular walls, mixing immiscible fluids, and dispersing nanomaterials.



Tungsten Halogen light Source:

A **Tungsten Halogen Light Source** (also known as a quartz-halogen or halogen lamp) is an advanced type of incandescent light source that produces a highly stable, continuous spectrum spanning from the **visible light range well into the near-infrared (NIR) spectrum** (typically from 360 nm to 2500 nm).

Because of its exceptionally smooth, drift-free output, it is widely utilized as a standard illumination source in optical spectroscopy, spectrophotometers, colorimeters, and medical diagnostic equipment.



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Spectrometer:

A **spectrometer** is a high-precision scientific instrument used to capture, split, and analyze the individual wavelengths of light or other electromagnetic radiation [wikipedia.org]. While an ordinary camera merges wavelengths to capture a composite color image, a spectrometer acts as an optical prism network. It breaks light down into its component colors (spectral trace) to reveal the chemical composition, temperature, and physical properties of a sample



Fume hood:

Fume Hood is a specialized engineering station optimized for handling hazardous chemicals used in lithography, etching, substrate cleaning, and wet benches. Unlike a standard chemistry fume hood, a fabrication lab hood is strictly integrated into a cleanroom environment, meaning it must protect both the operator from toxic vapors and the sensitive silicon wafers or substrates from airborne particle contamination.

